MCO-1D ORNITHOPTER

Bill of Materials — Revision 12

Complete parts list for the MCO-1D eight-wing metachronal ornithopter. Revision 12 reflects the complete software system as of July 2026: 42,115-line Wave Library, 23 active biomimetic hypotheses, full commissioning test procedure, Discovery Mode, active damping, precision descent controller, and water operations framework.

Document	Bill of Materials — Revision 12
Aircraft	MCO-1D Ornithopter
MTOW	3.448 kg
Date	July 2026
Software	Wave Library 42,115 lines
Hypotheses	23 active, 93 variants

1. AIRFRAME AND STRUCTURE

1	set	Main airframe / fuselage	Carbon fibre or composite structure. Mounts all hardware. Custom fabricated or modified from existing UAV frame.	Custom
4	pairs	Wing panels S1223 profile	Chord 165mm. Lightweight composite or film-covered foam. Must withstand flapping loads at 96° amplitude, 3Hz.	Custom
4	sets	Sector wheel assemblies	100° mechanical limit. One per wing pair. Mounts to sector wheel axle, driven by rope from motor drum.	Custom
4	sets	Rope drive assemblies	7:1 ratio. Motor drum to sector wheel drum. Inspect for fraying before every flight.	Cord/rope
8	pcs	Wing pivot hardware	Pitch servo horn to wing leading edge attachment. Must allow full AoA range without binding.	Hardware
1	set	Landing gear / float mounts	Skids for land ops. Float mounts for future water ops. Float specification TBD post-TC-02.	Custom

2. PROPULSION — MOTORS

4	pcs	GoolRC 5010-14 Brushless Motor	360 KV. 12N14P. 7 pole pairs. One per wing pair (drives both port and starboard). Verify: 5010-14 designation, NOT 5010-6 (different pole count).	~$25 each
4	pcs	Motor mount hardware	Bolts and standoffs for motor to airframe. Thread lock on all motor mount fasteners.	Hardware
4	pcs	Motor drum / pulley	Small drum on motor output shaft for rope drive. Sized for 7:1 ratio with sector wheel drum.	Custom

3. PROPULSION — ODRIVE CONTROLLERS

4	pcs	ODrive motor controller	ODrive v3.6 or v3.4S. CAN-FD capable. Node IDs 0,1,2,3 assigned to W1,W2,W3,W4. vel_limit must be set to 150 rad/s via mco_odrive_setup.py.	~$130 each

1	pcs	MCP2515 SPI-to-CAN adapter	Oscillator 8MHz. Interrupt on GPIO25 (or as wired). Connects Pi SPI0 to CAN-FD bus.	~$5
1	m	CAN bus cable	Twisted pair CAN-H and CAN-L. Daisy-chains ODrives 0→1→2→3.	Per length
2	pcs	CAN bus termination resistors	120 ohm, 0.25W minimum. One at each end of the bus. Required — bus will not work without terminators.	<$1

4. SERVOS — PITCH (ANGLE OF ATTACK)

8	pcs	TORQ TCM-1001 or equivalent digital servo	High-speed digital. 7.4V supply. 333Hz PWM capable. One per wing — controls angle of attack independently. GPIO pins: W1p=18, W1s=23, W2p=24, W2s=25, W3p=12, W3s=16, W4p=20, W4s=21.	~$15 each
1	set	Servo horns and linkage hardware	For all 8 pitch servos. Set to mechanical neutral before attaching to wing.	Hardware

5. SERVOS — SWEEP (FORE-AFT POSITION)

8	pcs	DS3225 or equivalent high-torque digital servo	25kg-cm torque, digital, 7.4V. Standard 50Hz PWM. One per wing — controls fore-aft wing position. GPIO pins: W1p=4, W1s=17, W2p=27, W2s=22, W3p=5, W3s=6, W4p=13, W4s=19.	~$10 each

6. FLIGHT COMPUTER

1	pcs	Raspberry Pi 4 Model B — 4GB RAM	4GB strongly recommended. 2GB minimum. Runs Wave Library at 50Hz. 4GB needed for extended discovery flights where registry grows large.	~$55
1	pcs	MicroSD card 64GB Class 10 or better	32GB minimum. 64GB recommended for extended logging. Flight logs at 50Hz fill approximately 50MB per hour.	~$10
1	pcs	Raspberry Pi case with cooling	Passive or active cooling. Motor vibration environment. Secure mounting — the Pi must not move during flight.	~$10

7. SENSORS

1	pcs	MPU-6050 IMU (I2C)	Or equivalent. 6-DOF inertial measurement unit. Provides roll, pitch, yaw rates at 100Hz. Primary attitude reference for all flight control.	~$3

1	pcs	BMP280 barometer (I2C)	Or equivalent. Provides altitude above launch point. Used above 3m where rangefinder loses reliability.	~$3
1	pcs	TF-Luna lidar rangefinder	Or equivalent 50Hz single-point lidar. Range 0.1-8m minimum. Used for landing height reference and water surface tracking in water operations.	~$20
1	pcs	DS18B20 temperature sensor (waterproof)	One-Wire bus on GPIO4 with 4.7kΩ pull-up to 3.3V. Outside air temperature for density altitude computation. Required for high-altitude SAR operations. Enable via: dtoverlay=w1-gpio,gpiopin=4 in /boot/config.txt.	~$3
1	pcs	HX711 load cell amplifier	DT pin: GPIO26. SCK pin: GPIO7. Used with strain gauge load cell for TC-01 and TC-02 static thrust measurement. Not mounted in flight.	~$2
1	pcs	Strain gauge load cell	Capacity: minimum 50N (5kg). For TC-01 thrust measurement bench fixture.	~$10

8. RADIO AND NAVIGATION

1	pcs	RC transmitter and receiver (SBUS or IBUS)	Minimum 4 channels. SBUS protocol to Pi UART. Ch1=Roll, Ch2=Pitch, Ch3=Lever, Ch4=Yaw. Verify channel mapping before TC-02.	Per system
1	pcs	Pixhawk flight controller	Acts as sensor bridge only — does not run flight control. Provides GPS, compass, additional IMU to Pi via MAVLink. Connected to Pi UART (TELEM2) at 921600 baud.	~$40
1	pcs	GPS module for Pixhawk	External GPS/compass. Mount away from motor noise. Used by Discovery Mode racetrack pattern.	~$20

9. POWER SYSTEM

1	pcs	Main LiPo battery — 4S, 4000mAh minimum	Powers motors through ODrives. 30C discharge rate minimum. Higher capacity for longer discovery flights.	Per spec
1	pcs	Servo LiPo battery — 2S, 1000-2000mAh	Dedicated supply for all 16 servos at 7.4V. Separate from main battery prevents motor noise on servo supply.	Per spec
1	pcs	BEC 5V 3A	Powers Raspberry Pi from main or servo battery. Must be a clean regulated supply — motor switching noise on Pi power can cause SD card corruption.	~$8
1	pcs	30A blade fuse and holder	Main battery lead fuse. Protects wiring on dead short.	~$3
1	pcs	5A blade fuse and holder	Pi power supply fuse.	~$2
1	pcs	Battery voltage monitor	Warns at 30% (amber) and 20% (red) remaining. Bus voltage read from ODrive telemetry in software.	~$5

10. PANEL AND DISPLAY

1	pcs	Touchscreen display — 3.5" or larger	SPI or HDMI. Runs mco-panel.service. Shows 4 buttons: PRE-FLIGHT, ARM, DISCOVERY, POST-FLIGHT. Plus six margin bars (altitude, stall, frequency, amplitude, gust, combined) in yellow/amber/red.	~$25

11. TOOLS AND CONSUMABLES

1	set	CAN bus test equipment	CAN analyzer or oscilloscope for bus verification. Verifies termination and signal integrity at TC-01.	Per shop
1	pcs	I2C bus scanner	Any I2C scanner to verify IMU and barometer addresses.	Software
1	pcs	Servo tester	Stand-alone servo tester for pre-installation servo check.	~$5
1	roll	Thread lock (Loctite Blue)	All motor mount fasteners. All sector wheel fasteners. Do not over-apply — makes future removal difficult.	Per roll
1	set	Conformal coating	For all PCBs if water operations are planned. Apply to Raspberry Pi, ODrive boards, and servo electronics.	Per can

12. SOFTWARE (NO COST)

1	file	mco_wave_library.py	42,115 lines. Main flight control. 23 active biomimetic hypotheses, 93 variants. Full commissioning, discovery, and advanced systems.	No cost
9	files	Supporting Python files	mco_main, mco_mavlink, mco_gs_server, mco_logger, mco_odrive_setup, mco_panel, mco_ground_station, mco_sync, mco_hypothesis_tool.	No cost
1	file	mco_loadcell.py	PENDING — HX711 load cell reader for TC-01/TC-02. To be written before first bench test.	No cost
1	pcs	Raspberry Pi OS Lite 64-bit	Downloaded from raspberrypi.com. Lite version (no desktop) for maximum control loop performance.	No cost

13. COMMISSIONING TEST EQUIPMENT

The following items are needed specifically for TC-01 and TC-02 and may not be permanently installed in the aircraft.

1	set	Load cell test fixture	Rigid frame to mount aircraft for static thrust measurement. Must withstand full thrust (>40N) without movement. HX711 + load cell + fixture together give thrust vs frequency curve.	Custom
1	set	Temperature measurement equipment	Infrared thermometer or thermal camera. Monitor motor temperature during TC-01-06 and TC-01-07.	~$20
1	pcs	Laptop for ground station	Runs mco_ground_station.py. WiFi to aircraft. Second operator during all TC flights.	Per availability
1	set	Field safety equipment	First aid kit. Fire extinguisher (LiPo fires). Safety glasses. Hearing protection for extended bench runs.	Per safety plan

14. SUMMARY AND SOFTWARE STATISTICS

MTOW	3.448 kg
Wings	8 (4 pairs W1-W4 fore to aft)

Motors	4 × GoolRC 5010-14 360KV
Servos	16 total (8 pitch + 8 sweep)
ODrives	4 (one per wing pair, CAN-FD bus)
Stall speed	4.85 m/s (9.4 knots) OGE
VNE	28.0 m/s (54.4 knots)
Frequency ceiling	3.0 Hz
Chord	165 mm
Re hover	~28,000
Re cruise	~89,000
Wave Library	42,115 lines Python
Total Python	~44,000 lines across 10 files
Active hypotheses	23 (H-01 to H-26 excluding H-09/10/11)
Hypothesis variants	93
Biological sources	11 animals
Failure combinations	255 pre-computed
Progressive chains	2,080 generated
Discovery focus modes	10
GS commands	98
Tick time Pi 4 estimate	~10ms mean (2× headroom)

✓ NOTE: Revision 12 reflects the complete software system. Hardware specification (float system for water operations) remains pending until hover capability is confirmed at TC-02 and float geometry is determined.